

Name: _____

These questions assess your understanding of the material from Chapters 3, 4 of Silberberg. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must explain why you chose your answer. Remember to report the units and correct number of significant figures in your answers.

- Andrew conducted some measurements and determined that 0.372 moles of vanadium participated in a chemical reaction. Calculate the mass of vanadium that participated in the chemical reaction.
ANSWER: _____
- Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reactant that is oxidized and the formula of the reactant that is reduced. Provide an explanation for why you made these choices.
 $\text{Rb} + \text{CdI}_2 \rightarrow \text{Cd} + \text{RbI}$
ANSWER: _____
- Calculate the mass of a sample of hydrogen monoiodide (HI) that contains 4.59×10^{15} iodine atoms.
ANSWER: _____
- An unknown chemical compound is composed of two unknown elements, A and B. The chemical formula is AB_x , where x represents a whole number. Measurements of a sample of the compound show that it contains 0.9025 moles of A and 3.61 moles of B. Write the complete chemical formula for the unknown compound.
ANSWER: _____
- Determine whether this reaction will occur. If it does, write the complete balanced equation. If it does not, write "no reaction occurs".
 $\text{Ca (s)} + \text{H}_2\text{O (l)} \rightarrow$
ANSWER: _____
- Given this unbalanced chemical equation, calculate and report the percent yield of Na_2S if 19.8 g of Na_2S is produced from the reaction of 31.37 g of Na_3PO_4 and 39.78 g of K_2S . Ensure that your answer has the correct number of significant figures.
 $\text{Na}_3\text{PO}_4 + \text{K}_2\text{S} \rightarrow \text{K}_3\text{PO}_4 + \text{Na}_2\text{S}$
ANSWER: _____
- Lead (II) hydroxide reacts with hydrochloric acid to form water and lead (II) chloride. Suppose 186.9 g of lead (II) hydroxide is mixed with 43.61 g of hydrochloric acid. Calculate the maximum mass of water that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.
ANSWER: _____
- The extraction of zinc from its sulfide ores involves a two-step roasting and reduction process. In the first step, zinc sulfide reacts with oxygen to form zinc oxide and sulfur dioxide. In the second step, zinc oxide is reduced by carbon to form zinc metal and carbon monoxide. Write the net chemical equation for the production of zinc from zinc sulfide, oxygen, and carbon. Ensure that it is balanced.
ANSWER: _____
- Which element is 4% by mass of triiodine mononitride (NI_3)? Show your work.
ANSWER: _____

10. To help farmers and gardeners, commercial fertilizers have a big three-number "NPK" label on the bag that gives the amounts of three key plant nutrients: nitrogen, phosphorus, and potassium. For example, a bag of "10-10-10" fertilizer has 10% by mass nitrogen, 10% by mass potassium as potash (K_2O) and 10% by mass phosphorus as phosphate (P_2O_5).

ingredient	percent by mass
calcium nitrate ($Ca(NO_3)_2$)	17.0%
urea ($(NH_2)_2CO$)	18.0%
potassium sulfate (K_2SO_4)	13.0%
ammonium phosphate ($(NH_4)_3PO_4$)	24.0%
inert ingredients	28.0%

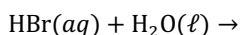
What should the first (nitrogen) number on the bag's label be? In other words, calculate the percent by mass of nitrogen in this fertilizer. Round your answer to the nearest tenth of a percent.

ANSWER: _____

11. A virus has a mass of 3.38×10^{-11} mg and a brick has a mass of 9.14×10^0 kg. What is the mass of 1 mole of viruses? Ensure your answer has the correct number of significant figures.

ANSWER: _____

12. Predict the products of the reaction below by completing the right-hand side of the chemical equation. Be sure your equation is balanced and contains state symbols after every reactant and product.



ANSWER: _____

13. Given this unbalanced chemical equation, report the amount of $ZnCrO_4$ needed to produce 1.44 moles of ZnI_2 . Report your answer in moles. Ensure your answer has the correct number of significant figures.
- $$LiI + ZnCrO_4 \rightarrow ZnI_2 + Li_2CrO_4$$

ANSWER: _____

14. Natalie weighs out 1.461 g of an unknown triprotic acid into a 100.-mL volumetric flask and dilutes the solution to the mark with distilled water. She then titrates this solution with 0.336-M sodium hydroxide solution. When the titration reaches the equivalence point, Natalie finds that she has added 133.12 mL of the sodium hydroxide solution. Calculate the molar mass of the unknown triprotic acid. Ensure your answer has the correct number of significant figures.

ANSWER: _____

15. Jennifer must dilute 36.66 mL of a 4.465-mM solution of aqueous ammonium oxide ($(NH_4)_2O$) until the concentration falls to 1.341 mM. She will do this by adding distilled water to the solution. How much distilled water must she add? Be sure your answer has the correct number of significant figures.

ANSWER: _____

16. Nathan puts a small sample of K_2O in water and lets it dissolve. After equilibrium is reached, he determines that several chemical species remain in the solution. These include K^+ , O^{2-} , and H_2O . Which of the following labels apply to the compound K_2O ? Write all that apply. Possible labels include ionic, molecular, strong acid, weak acid, strong base, and weak base. Explain why you chose your answer.

ANSWER: _____

17. Write the net ionic equation for the following chemical reaction, which takes place in aqueous solution:
- $$NH_4F + AgBrO \rightarrow AgF + NH_4BrO$$

ANSWER: _____

18. Nitric acid reacts with rubidium hydroxide to form water and rubidium nitrate. Calculate the moles of rubidium hydroxide needed to produce 1.21 moles of water. Ensure your answer has the correct number of significant figures.

ANSWER: _____

19. Samantha weighs out 1.841 g of boric acid (H_3BO_3) into a 100.-mL volumetric flask and dilutes the solution to the mark with distilled water. She plans to titrate the acid with 0.3147-M strontium hydroxide solution. Calculate the volume of strontium hydroxide solution that Samantha must add to reach the final equivalence point. Ensure your answer has the correct number of significant figures.

ANSWER: _____

20. Jennifer conducted some measurements and determined that 0.169 moles of iodine solid was produced in a chemical reaction. Calculate the mass of iodine solid that was produced in the chemical reaction.

ANSWER: _____

21. Andrew is performing a combustion analysis on a compound with a molecular molar mass of 130.23 g/mol that is composed of carbon, hydrogen, and perhaps oxygen. A sample of 6.19 g of the compound is completely burned in excess oxygen and the mass of the products is carefully measured:

Product	Mass (g)
Carbon Dioxide	16.73 g
Water	7.71 g

Use this information to find the molecular formula of the compound.

ANSWER: _____

22. Nitric acid is often manufactured in a two-step process from the atmospheric gases nitrogen and oxygen plus hydrogen, prepared by reforming of natural gas. In the first step, nitrogen and hydrogen react to form ammonia. In the second step, ammonia and oxygen react to form nitric acid and water. Write the net chemical equation for the production of nitric acid from nitrogen, hydrogen, and oxygen. Ensure that it is balanced.

ANSWER: _____

23. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 57.5 g.

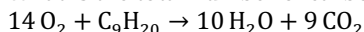
Compound	Initial Mass	Final Mass
A	56.4 g	38.4 g
B	71.9 g	0 g

ANSWER: _____

24. Determine which of the following labels apply to the chemical compound Si_3Sb_4 : ionic, molecular, acid, and base.

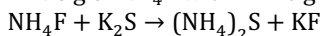
ANSWER: _____

25. What is the total number of carbon atoms on the right-hand side of this chemical equation?



ANSWER: _____

26. Given this unbalanced chemical equation, report the maximum mass of $(\text{NH}_4)_2\text{S}$ that could be produced by the mixing of 11.98 g of NH_4F with 11.10 g of K_2S . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

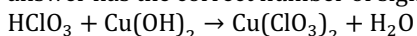
27. Calculate the number of germanium atoms in a 358.-g sample of trigermanium tetranitride (Ge_3N_4).

ANSWER: _____

28. Suppose Benjamin heats a sample of a certain aluminum sulfide hydrate, $\text{Al}_2\text{S}_3 \cdot 6\text{H}_2\text{O}$ until all the water is removed and pure Al_2S_3 is left behind. Find the percent by which the mass of the sample decreases in this process.

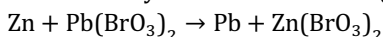
ANSWER: _____

29. Given this unbalanced chemical equation, report the mass of HClO_3 needed to produce 13.02 g of H_2O . Ensure your answer has the correct number of significant figures.



ANSWER: _____

30. As $\text{Pb}(\text{BrO}_3)_2$ is converted into Pb in the following reaction, determine whether lead is oxidized, reduced, or neither. Consider only the lead found in $\text{Pb}(\text{BrO}_3)_2$ and Pb.



ANSWER: _____

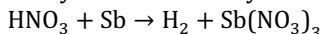
31. Manganese commonly occurs in nature as a mineral. The extraction of manganese from the carbonite mineral rhodochrosite involves a two-step process. In the first step, manganese (II) carbonate and oxygen react to form manganese (IV) oxide and carbon dioxide. In the second step, manganese (IV) oxide and aluminum react to form manganese and aluminum oxide. Write the net chemical equation for the production of manganese from manganese (II) carbonate, oxygen, and aluminum. Ensure that it is balanced.

ANSWER: _____

32. Samantha conducted some measurements and determined that 1.3897 g of tellurium participated in a chemical reaction. Calculate the number of moles of tellurium that participated in the chemical reaction.

ANSWER: _____

33. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). If you believe that the reaction cannot be classified as one of these types, write how you would classify it.



ANSWER: _____

34. Lauren needs to make 453.10 mL of aqueous ammonium oxide $((\text{NH}_4)_2\text{O})$ working solution. She plans to do this by adding distilled water to a 90.0-mL stock solution of 0.419-M aqueous ammonium oxide. What will be the concentration of her working solution? Report your answer in mM. Be sure your answer has the correct number of significant figures.

ANSWER: _____

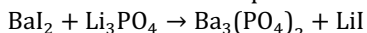
35. A brick has a mass of 7.89×10^0 kg and a cooking pot has a mass of 6.324×10^{-1} kg. What is the mass of 1 mole of cooking pots? Ensure your answer has the correct number of significant figures.

ANSWER: _____

36. Hydroiodic acid reacts with potassium hydroxide to form water and potassium iodide. Calculate the mass of hydroiodic acid needed to produce 15.95 g of potassium iodide. Ensure your answer has the correct number of significant figures.

ANSWER: _____

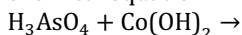
37. A way to produce solid barium phosphate is by combining aqueous barium iodide with aqueous lithium phosphate. The chemical reaction is represented by the following unbalanced chemical equation:



Jason wants to produce 29.54 mg of barium phosphate. He pours 10.63 mL of a 0.0134-M solution of aqueous barium iodide into a beaker. What volume of a 0.09920-M solution of aqueous lithium phosphate is needed to react with the barium iodide solution to produce 29.54 mg of barium phosphate? If you find that 10.63 mL of the barium iodide solution is insufficient, report the required volumes of both reactants required to produce the 29.54 mg of barium phosphate.

ANSWER: _____

38. Predict the products of the reaction below. In other words, write the products to complete the right-hand side of the chemical equation. Ensure the final equation is balanced.



ANSWER: _____

39. The chemical formula for diphosphorus trioxide is P_2O_3 . Calculate the molar mass of diphosphorus trioxide. Round your answer to 4 significant figures.

ANSWER: _____

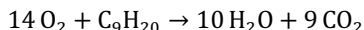
40. Hannah prepares a solution of hydronium sulfide $((\text{H}_3\text{O})_2\text{S})$ by measuring out 83.1 mmol of hydronium sulfide into a 100.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

41. Shiv adds 123.14 mL of a 0.214-M solution of caesium phosphate (Cs_3PO_4) to a reaction flask. Calculate the amount of caesium phosphate that he has added to the flask. Report your answer in moles and be sure your answer has the correct number of significant figures.

ANSWER: _____

42. What is the total number of carbon atoms on the right-hand side of this chemical equation?



ANSWER: _____

43. Titanium is extracted from its ore, rutile (titanium dioxide), via the two-step Kroll process. In the first step, titanium dioxide is converted to titanium tetrachloride by reacting it with chlorine and carbon. In the second step, titanium tetrachloride is reduced using molten magnesium to form titanium metal and magnesium chloride (which is then often refined back to magnesium). Write the net chemical equation for the production of titanium from titanium dioxide, chlorine, carbon, and magnesium. Ensure that it is balanced.

ANSWER: _____

44. The extraction of aluminum metal from the aluminum hydroxide in bauxite ore by the Hall-Héroult process is one of the most remarkable success stories of late-19th-century chemistry, turning aluminum from a rare and precious metal into the cheap commodity it is today. In the first step, aluminum hydroxide reacts to form alumina (which is aluminum oxide) and water. In the second step, alumina and carbon react to form aluminum and carbon dioxide. Write the net chemical equation for the production of aluminum from aluminum hydroxide and carbon. Ensure that it is balanced.

ANSWER: _____

45. What major chemical species are present in an aqueous solution of 0.014-M aluminum phosphate?

ANSWER: _____

46. Measurements show that an unknown compound has the following composition:

Element	Mass %
B	43.56 %
N	56.44 %

Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

ANSWER: _____

47. A strip of solid strontium metal is put into a beaker of 0.080-M LiBrO_3 solution. Does a chemical reaction occur? Explain why it does or why it does not. If it does occur, write the net ionic equation that describes it. Be sure your equation is balanced and has physical-state symbols.

ANSWER: _____

48. Iron is extracted from its oxide ores in a two-step process that has been practiced for centuries. In the first step, iron (III) oxide reacts with carbon monoxide to produce iron (II) oxide and carbon dioxide. In the second step, iron (II) oxide further reacts with carbon monoxide to produce elemental iron and more carbon dioxide. Write the net chemical equation for the production of iron from iron (III) oxide and carbon monoxide. Ensure that it is balanced.

ANSWER: _____

49. Calcium hydroxide reacts with hydrosulfuric acid to form calcium sulfide and water. If 4.09 g of calcium sulfide is produced from the reaction of 6.44 g of calcium hydroxide and 4.101 g of hydrosulfuric acid, calculate the percent yield of calcium sulfide. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

50. Tyler measures the amount of silicon dipolonide produced during a chemical reaction. He finds that 0.20 mol is produced. Calculate the mass of silicon dipolonide produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____